

AI Engineering: Fast-Track Internship Syllabus

Goal: Rapid foundational engineering training and end-to-end deployment of a live AI production script or pipeline to meet mandatory college placement credits.

Format: Project-Based Fast-Track Deployment + Weekly Troubleshooting Syncs • **Duration:** 2.5 Weeks (10 Intense Classes)

Week 1 Sprints: Syntactical Mechanics & Ingestion Logic (Classes 1-5)

Focus: Mastering programmatic prompt design rules, eliminating hallucinations, and setting structural pipeline constraints.

Session	Topic	Core Learning Objectives & Details
Class 1	Ecosystem Architecture & Advanced Prompting	Core Focus: Transitioning from simple browser chat prompts to structural role-setting constraints. Mastering the Role, Context, Task, and Constraints (RCTC) text framework. Required Milestone: A formalized multi-layered markdown instruction structure that produces zero conversational fluff when evaluating sample text strings.
Class 2	Structured Syntactical Tagging & Data Isolation	Core Focus: Implementing deep structural XML tagging sequences (<context>, <rules>, <output_schema>) to isolate input data fields and prevent context bleeding. Required Milestone: An operational parsing prompt that successfully extracts targeted text segments and matches them to clean layout wrappers.
Class 3	Conditional Logic Strings & Validation Gates	Core Focus: Designing programmatic "If/Then/Else" instruction sequences to eliminate model hallucinations and gracefully catch out-of-bounds inputs. Required Milestone: A validation prompt that scans incoming logs, flags missing components, and forces the model to stop and output explicit errors.

Session	Topic	Core Learning Objectives & Details
Class 4	Prompt Chaining Frameworks	Core Focus: Breaking down complex data tasks into serialized steps where the precise text output of Stage A automatically becomes the raw input for Stage B. Required Milestone: A functioning 2-stage manual prompt sequence that ingests raw copy, summarizes it, and passes the clean output into a format engine.
Class 5	Programmatic API Mechanics & Payload Construction	Core Focus: Understanding how code interactions bridge with foundation cloud models. Deconstructing HTTP request headers, variables (temperature, top_p), and raw JSON response objects. Required Milestone: A complete configuration chart mapping out how a local script authenticates with and parses data from a cloud model gateway safely.

Week 2 Sprints: Application Engineering & Project Defense (Classes 6-10)

Focus: Stitching component layers into a running application backend, error hardening, and terminal project defense.

Session	Topic	Core Learning Objectives & Details
Class 6	Live Capstone Project Selection & Scoping	Core Focus: Evaluating real-world deployment tracks based on your website's focus area (e.g., Automated Resume Tracker, Medical Insights Extraction Assistant, or Smart Document Matrix). Required Milestone: A confirmed project roadmap and system block-flow diagram formally signed off by the internship coordinator.

Session	Topic	Core Learning Objectives & Details
Class 7	Building the System Integration Blueprint	Core Focus: Code integration day. Stitching Class 2 tag boundaries, Class 3 validation checks, and Class 4 sequence chains into a single unified pipeline execution file. Required Milestone: A working backend terminal script executing all core processing blocks concurrently against sample datasets.
Class 8	Edge Case Remediation & Output Hardening	Core Focus: Subjecting system execution loops to stress testing. Ingesting intentionally corrupted text records, massive files, or empty parameters to implement runtime safety blocks. Required Milestone: An updated codebase incorporating functional error handling and strict output schema constraints.
Class 9	UI Integration Sandbox & Local Hosting Showcase	Core Focus: Wrapping back-end logic processing inside browser-executable no-code sandboxes or basic local interfaces for user-friendly testing. Required Milestone: A working visual preview wrapper displaying real-time text generations based on dynamic slider inputs.
Class 10	Technical Defense Presentation & Certificate Sign-Off	Core Focus: Technical review panel. Interns present their live running system, defend their pipeline choices, and demonstrate architectural stability. Required Milestone: Presentation completion, code repository upload, and formal issuance of the AI Placement Completion Certificate for university records.